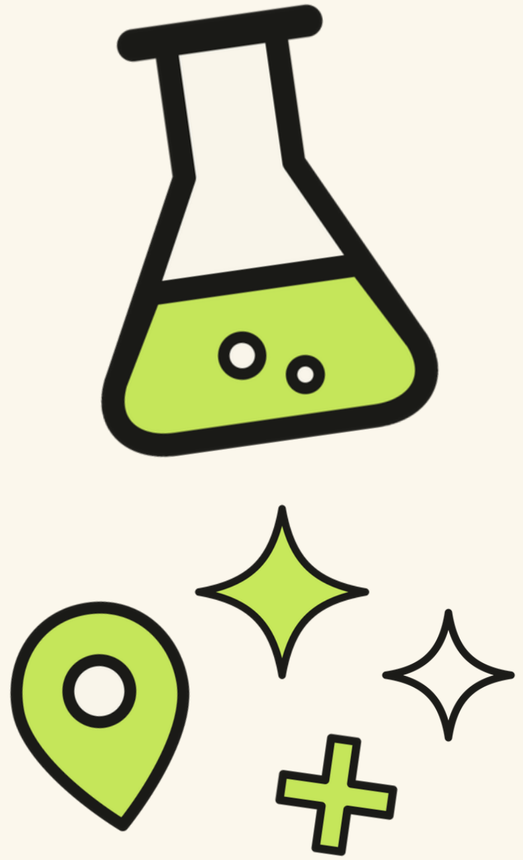
WORKSHOP #6

SCIENTIFIC WRITING

PART 2

READING, INTERPRETING, AND SUMMARIZING DATA



STATE IT

SHOW IT

EXPLAIN IT

DEBORAH TORRICO-PARDO & DEV SRIVASTAVA

NEXTINRESEARCH 2026

CHECK-IN / TURN AND TALK:

TAKE ONE MINUTE, THEN SHARE:

- WHERE ARE YOU IN YOUR RESEARCH PROCESS?
- WHAT DATA OR OBSERVATIONS DO YOU HAVE SO FAR?
- WHAT IS HARDEST TO EXPLAIN CLEARLY?
- FOR GROUPS: ARE YOU INTERPRETING THE DATA IN THE SAME WAY?

AND, WHAT QUESTIONS DO YOU HAVE FOR US?

TODAY'S LEARNING GOALS:

1

READ

Return to your question, hypothesis, and methods.

2

STATE

Write a neutral result sentence with the needed statistics.

3

SHOW

Choose a figure that makes the relationship easy to see.

4

DISCUSS

Explain patterns, limitations, and possible meaning.

REVIEW FROM LAST MEETING: RESULTS VS. DISCUSSION

RESULTS = WHAT

- **Summarize data** in text, tables, and figures
- **Report** significant and non-significant outcomes
- **Link** each finding to the research question
- **Do not** explain why yet

DISCUSSION = WHY / SO WHAT

- **Interpret** patterns and unexpected findings
- **Compare findings** w/ literature review
- **Explain** limitations and alternatives
- **Describe implications** + next steps

TURN AND TALK (REVIEW!): WHICH ONE DOESN'T BELONG?



A THE TREATMENT GROUP HAD A LOWER AVERAGE SCORE THAN THE CONTROL GROUP.

B THE PATTERN MAY REFLECT DIFFERENCES IN PARTICIPANT EXPERIENCE.

C THE SIGNIFICANT DIFFERENCE IN SLEEP BETWEEN GROUP A AND B SUGGESTS COFFEE AFFECTS SLEEP.

WHAT WORD OR PHRASE GAVE YOU THE CLUE?

RESULTS:

- 1. WORKING WITH RAW RESULTS
- 2. WRITING RESULTS STATEMENTS AND SUBSECTIONS
- 3. POWER OF FIGURES



1. HOW TO WORK WITH RAW RESULTS:

1. REVISIT YOUR GOAL (HYPOTHESIS)

2. REVISIT YOUR METHODS

3. FIND PATTERNS

HOW TO WORK WITH RAW RESULTS



1. REVISIT YOUR GOAL (HYPOTHESIS)

- What are you trying to prove?
- What results would prove this true?

HOW TO WORK WITH RAW RESULTS



2. REVISIT YOUR METHODS

- Understand how you got your result
- What units will your result be in? What are extreme values for that unit?
- Ex 1: If you are finding sleep time, then 20 hours is an extreme by context

HOW TO WORK WITH RAW RESULTS



3. FIND PATTERNS

- Ex 2: Hypothesize that a gene is present in only infected samples, so we tested both infected and control DNA for that gene
- The RESULT that would support this would have the gene expressed in the infected but not control



2. WRITING RESULTS STATEMENTS AND SUBSECTIONS

1. TITLE THE RESULT

2. STATE WHAT YOU ARE DOING

3. GIVE THE RESULT + SIGNIFICANCE

WRITING RESULTS STATEMENTS AND SUBSECTIONS

1. TITLE THE RESULT

- Title each subsection with each result that you found
- **Often multiple even if they are only tangentially/minimally related to your hypothesis alone**
- **Ex**: For the gene-in-infections paper, even though we are mainly focused on the expression across infected and control DNA, another result may also be **time of infection**. Could this create further evidence for your overarching goal?

TURN AND TALK:



LOOK AT YOUR STUDY'S MAIN HYPOTHESIS.

WITH YOUR NEIGHBOR, BRAINSTORM 2–3 SPECIFIC SUB-RESULTS OR SECONDARY MEASUREMENTS ASIDE FROM WHAT YOU ALREADY PLANNED ON RECORDING, THAT YOU COULD COLLECT

WRITING RESULTS STATEMENTS AND SUBSECTIONS

2.STATE WHAT YOU ARE DOING

- List out comparison groups and any statistical test used as your first few sentences
- List **what** from the groups you're comparing
- Include sample size (n= a number) beside the group if applicable
- Ex: control group of high schoolers (n=20) and high schoolers who play video games (n=25)

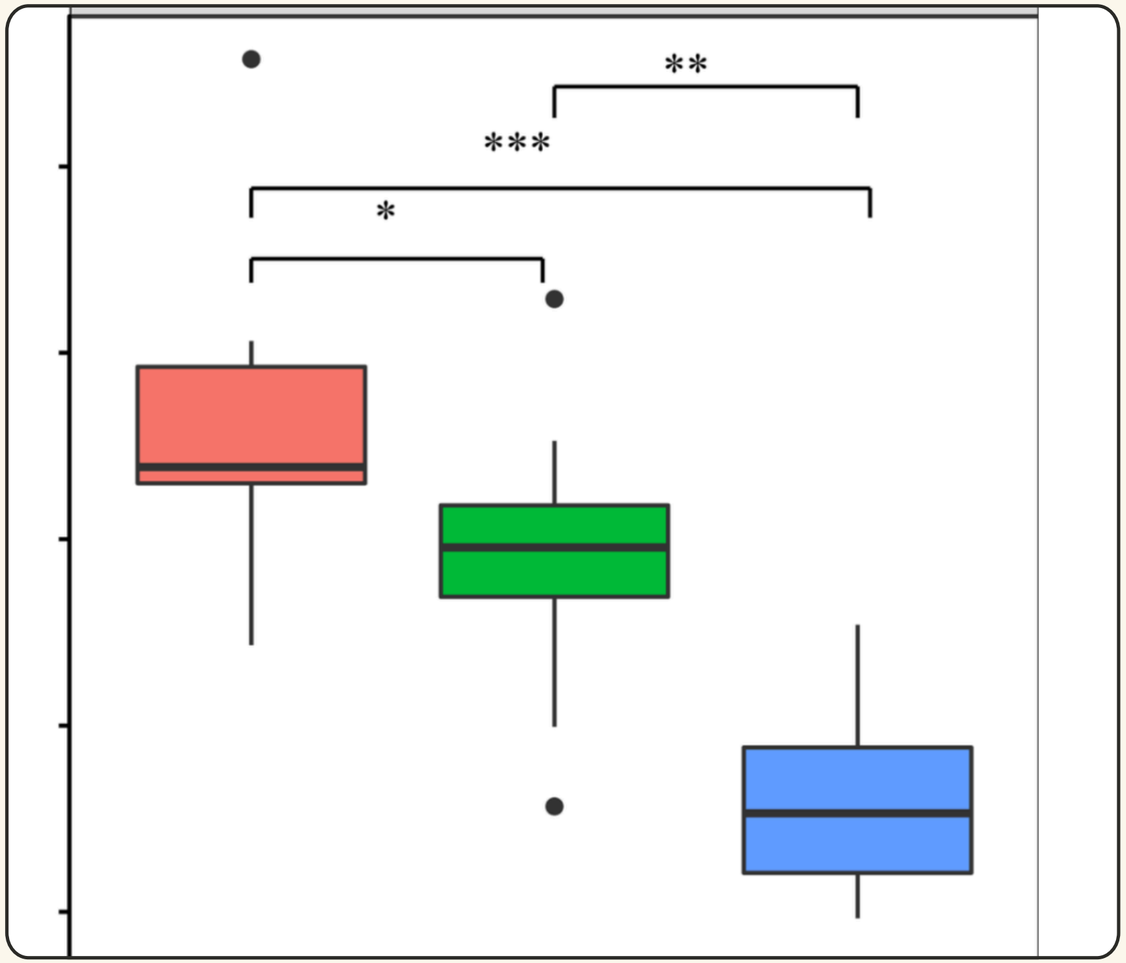
WRITING RESULTS STATEMENTS AND SUBSECTIONS

3. GIVE THE RESULT + SIGNIFICANCE

- List out your analyzed data
- State whether or not the result is significant
- Even if your result is not above the threshold of either p=0.01 or p=0.05, then you should still report the data!
- (Thresholds for p=0.01 vs. p=0.05 depend on context + “how sure you want to be”

3. WHY GRAPHS?

TURN AND TALK:

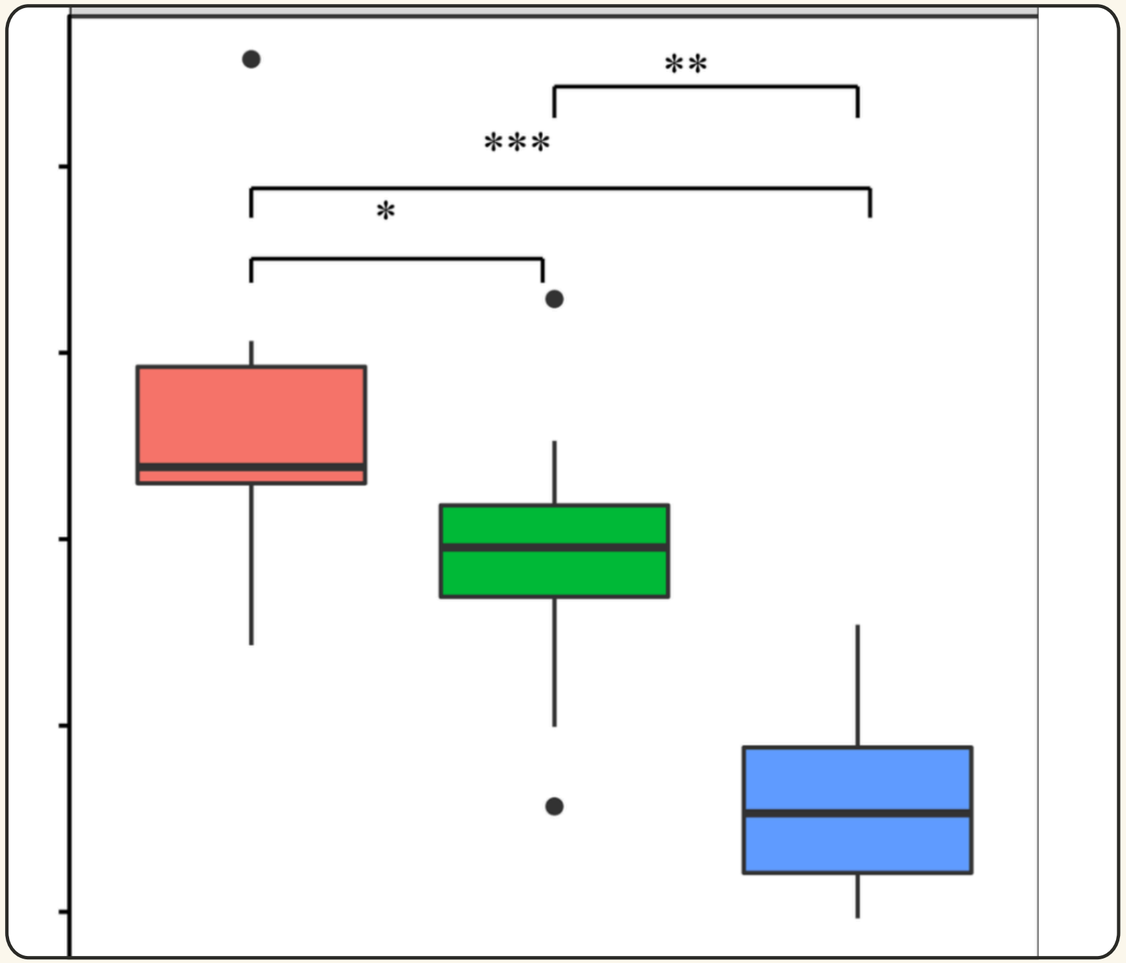


USING ONLY WORDS, SAY ALL THE RELATIONSHIPS GIVEN BY THIS GRAPH.

CALL EACH COLOR "GROUP RED, GROUP GREEN, AND GROUP BLUE"

NOTE: THE STARS INDICATE SIGNIFANCE, AND MULTIPLE STARS INDICATE SIGNIFIANCE AT LOWER THRESHOLDS (0.05, 0.01, 0.001)

TURN AND TALK:



KEY TAKEAWAY: THE VISUAL ALONE CAN SAY A LOT MORE, A LOT QUICKER!

(IMAGINE IF INSTEAD OF GROUP [COLOR], THEY HAD DENSE SCIENTIFIC NAMES!)

IMPLEMENTING GRAPHS

CAPTIONS

- When you give a result, it is often very helpful to refer the reader to a figure!
- When you include a figure, you will also have a caption with Fig. # [what your graph displays] and then information about the contents of the graph (brief summary)
- cite in-text using figure/table #

IMPLEMENTING GRAPHS

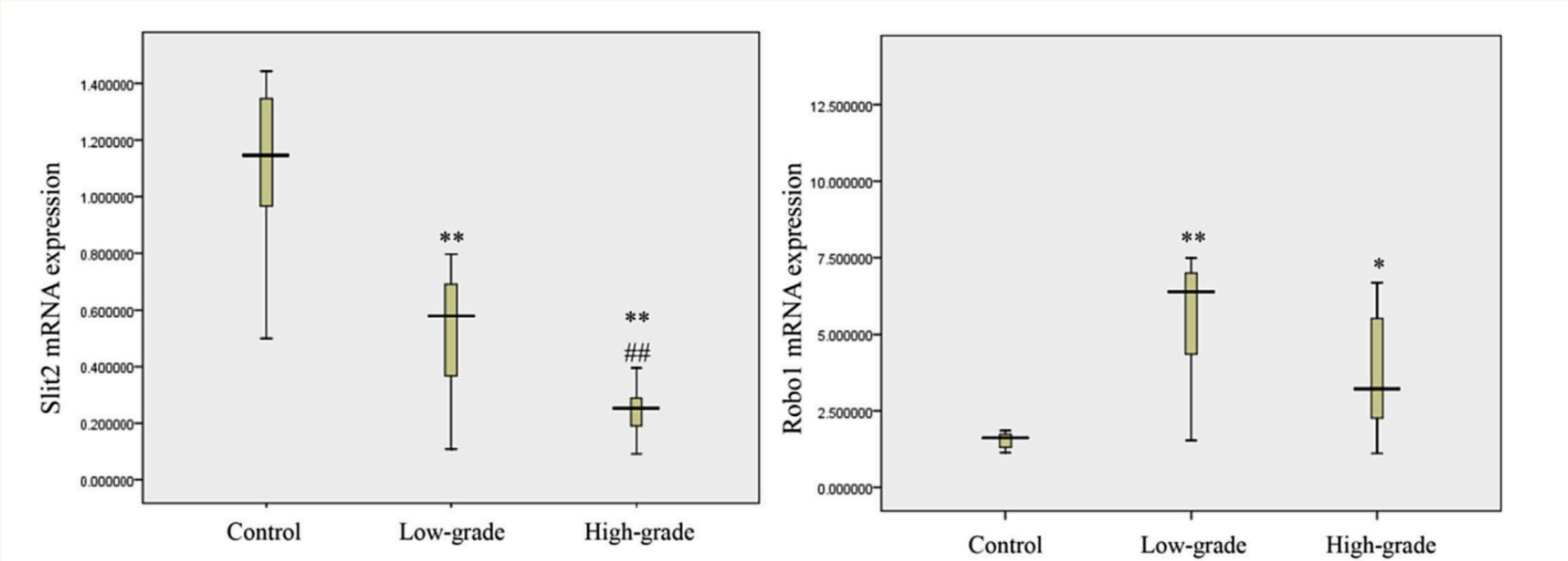
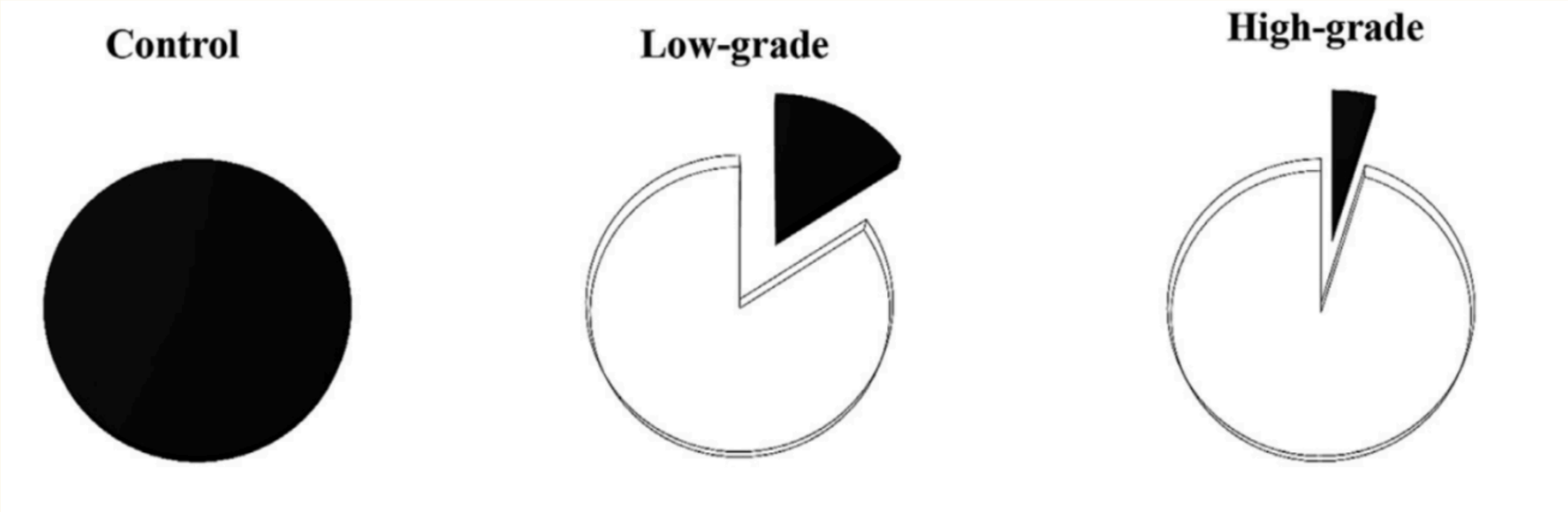


Fig. 3. Slit2 and Robo1 mRNA expression levels. Slit2 and Robo1 mRNA expression was examined in control, low- and high-grade glioma cells. The data were expressed as a fold change in the mRNA expression level of the individual tissues. The Slit2 and Robo1 mRNA levels were calculated in reference to the control mRNA level. The results were shown as median (IQR). *, **: a p-value of <0.05 or 0.01 vs the control. ##: a p-value of <0.01 vs low-grade gliomas.

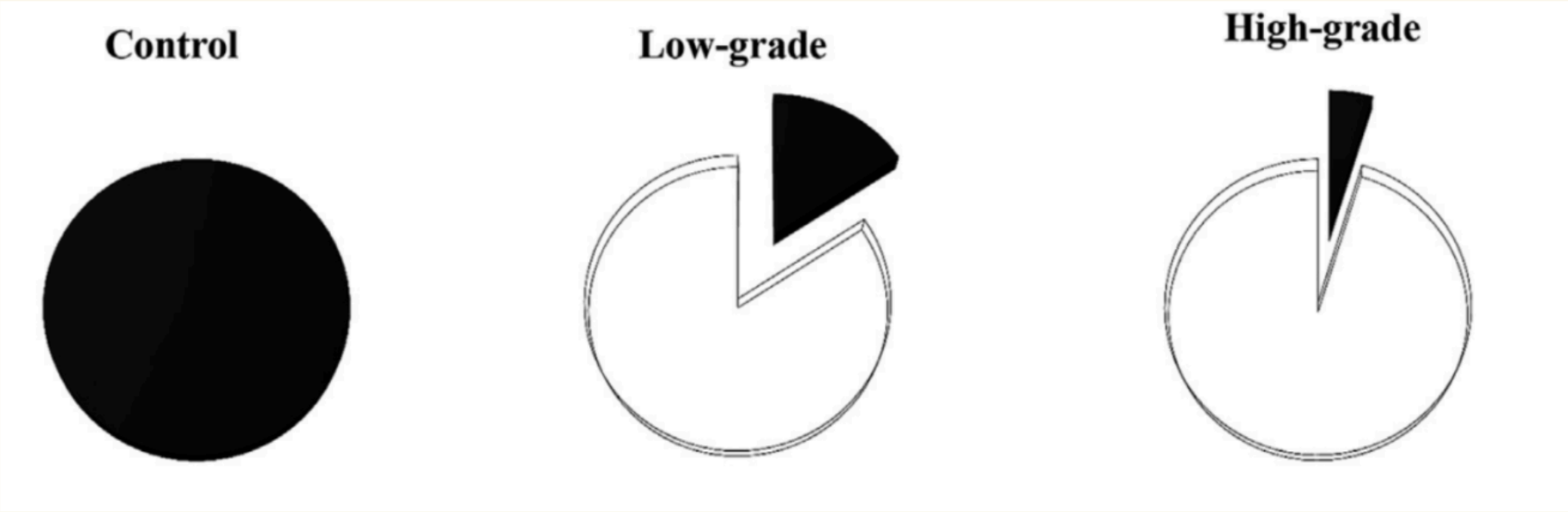
CHECK: AXES AND UNITS, COMPARISON GROUPS, DIRECTION OF THE PATTERN, CENTRAL VALUE AND VARIABILITY, SIGNIFICANCE MARKERS, SAMPLE SIZE AND CAPTION DETAILS

TURN AND TALK:

LOOK AT THIS GRAPH. WHEN WOULD THIS BE A GOOD GRAPH TO USE VERSUS NOT? WHAT COULD BE MISLEADING ABOUT THIS GRAPH?



NOTE: BE SURE TO SELECT THE CORRECT GRAPH!



^ HERE, THIS MAY BE A GOOD GRPAH TO SEE A GENERAL RELATIONSHIP. BUT IF YOU ARE INTERESTED IN SPECIFIC NUMERICAL DIFFERENCES, THIS IMAGE ALONE DOES NOTHING!

IS THE SHADED REGION IN HIGH-GRADE 10%? 5%? 3%? UNKNOWN!

USING GRAPHS WILL HELP YOU ALSO WITH WRITING THE DISCUSSION, WHERE YOU WILL HAVE TO MAKE CONCLUSIONS ABOUT YOUR DATA!

BRIEF DISCUSSION OVERVIEW:

- In the discussion, you will have to talk about the **spotted trends & patterns**.
- You can refer back to your graphs from the results for this!
- If you see something out of the ordinary, try to come up with a **narrative** to explain why this result may be
    - (also reference your literature review articles. did you get a **different result than them**? why?)

FUN ACTIVITY/BREAK:

- Any surveys that need to be completed?
- Ask for help! Plenty of mentors today, so if you have any questions about anything, please ask them now!